



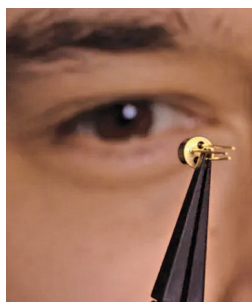
CPU cores, and I/O on a single chip, reducing system size and the bill of materials.

Ambarella

<https://www.ambarella.com>

Avalanche photodiode infrared sensors

Many IR systems struggle with limited sensitivity, which can reduce range, accuracy, and the ability to detect



small or low-reflectivity targets. Phlux Technology's Aura Noiseless InGaAs APDs address this by providing about 12 times the sensitivity

of standard InGaAs APDs, enabling systems to reach longer distances or use lower laser power, which also reduces cost, size, and weight and eases thermal design. They are suited for engineers and designers working on laser range finders, OTDRs, long-range lidar, free-space optical links, and hollow-core fibre systems, particularly where eye-safe operation, high dynamic range, fast recovery from reflections, and reliable performance up to +85°C are needed.

Phlux Technology

<https://phluxtechnology.com>

Inertial measurement unit

Smart wearables and hearables require motion-tracking sensors that use minimal power, and Bosch



Sensortec's BMI423 IMU (inertial measurement unit) meets this need by combining a wide measurement range with low power consumption.

It captures motion data from devices such as wearables, earbuds, fitness trackers, and robots while allowing the main processor to sleep to save battery. The BMI423's accelerometer measures $\pm 32g$, and its gyroscope ± 4000 degrees per second, doubling the range of the previous generation, enabling it to track fast movements that standard sensors often miss, making it suitable for sports tracking, gaming, or robotic control. On-sensor features such as bone-conduction voice detection activate microphones only when the user speaks, saving power and protecting privacy, while gesture recognition, multi-tap detection, and step counting are handled on the sensor, reducing processor load and extending battery life in devices.

Bosch Sensortec

<https://www.bosch-sensortec.com>

Single-chip LCOS panel

OMNIVISION has launched the OP03021, the industry's only single-chip liquid crystal on silicon (LCOS) panel with ultra-low power for next-generation smart glasses. The 0.26-inch (6.6mm) LCOS microdisplay



delivers 1632×1536 resolution at 90Hz, enabling higher resolution and a wider field of view to support a more immersive and comfortable augmented reality (AR) experience. The panel integrates a CMOS LCOS cell, an ASIC driver, frame buffer memory, and a MIPI receiver on the silicon backplane, reducing both size and power consumption, which are key factors for wearable designs.

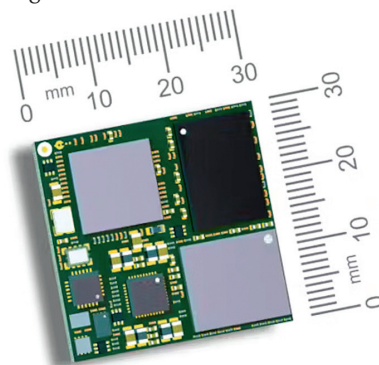
Omnivision

<https://www.ovt.com>

COM families

Many industrial automation and IoT devices struggle with limited space, difficult servicing, and the need for reliable, long-term edge computing,

especially when AI or real-time processing is required. Toradex's OSM and Lino computer-on-module (COM) families address these challenges. OSM's 30mm $\times$ 30mm soldera-



ble BGA design fits into tight spaces for permanent, rugged installations, while Lino's connector-based modules make upgrades and servicing easy. Both offer flexible hardware. Together, these COMs enable developers to build compact, intelligent, and maintainable devices efficiently.

Toradex

<https://www.toradex.com>

Solid-state active cooling chip

Devices running AI Reasoning Models, handheld gaming, storage, or 5G/Wi-Fi radios often overheat, limiting



performance and causing slowdowns, errors, or shorter lifespans. The AirJet Mini G2 from Frore Systems solves this by removing 50% more heat within the same footprint, allowing notebooks, tablets, mini-PCs, smartphones, cameras, SSDs, and industrial or IoT devices to run without thermal throttling. Its design keeps devices running under sustained workloads, improving performance and stability across consumer, industrial, datacentre, defence, and automotive applications.

Frore Systems

<https://www.froresystems.com>